

LUKE K. KIM

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SUMMARY

Multidisciplinary product development engineer taking consumer electronics from concept through high-volume overseas production across mechanical, electrical, and software. Rapid prototyping, injection-molded part CAD refined with ODM tooling partners, custom PCB design and embedded firmware, and root cause analysis. Certified SolidWorks Associate (CSWA).

EXPERIENCE

COCCO Hair Professional | Cleveland, OH
Product Development Engineer

June 2024 – Present

- **Improved blade-gap calibration precision 10x** (± 0.05 to ± 0.005 mm) and **cut cycle time 80%** by building a robotic fixture spanning **mechanical design, computer vision, and closed-loop motor control**; adopted as a factory calibration and QC step
- Direct product development across trimmer, clipper, and dryer programs from concept through manufacturing release as first point of contact for technical issues; newest product ships **~20,000 units/month**
- Developing a trimmer/clipper control PCB from the ground up (Altium Designer, ESP32 firmware); built factory proof-of-concepts including a miniaturized analog tachometer clipper (stepper-driven needle, adaptive engine audio, BLDC drive) **widely considered infeasible at that size**
- Owned a new product **end-to-end**: consumer research, sourcing, testing, product and packaging design, production launch
- Cut **PCB-related warranty claims by 84%** and **switch claims by 55%** across 57,000+ units shipped by owning failure analysis between the US team and the overseas factory, isolating the no-power mode to firmware rather than hardware, and redesigning high-failure switch mechanisms and charging contacts with tooling engineers through injection-mold DFM
- Led development of a novel low-vibration motor on-site at a supplier's factory in China, achieving **~5x lower vibration** than baseline in a prototype queued for launch; conduct line audits and first-article inspections on repeated visits
- Built standardized 3D-printed test adapters for airflow, thermal, vibration, and sound-level measurement to **benchmark products against competitors**, informing noise-reduction and blade-selection decisions
- Automated internal reporting (**1–2 hr/day**) and lead an internal team building AI-driven software tools on two-week sprints

Cambridge Mobile Telematics | Boston, MA
Mechanical Engineering Intern, IoT Devices

June 2023 – August 2023

- Designed IoT product components, manufacturing fixtures, and 3D-printed test housings in Onshape, improving assembly efficiency and streamlining testing during material changes
- Built a battery charge/discharge test PCB (Altium, Arduino); resolved field and line failures through root cause analysis

Vanderbilt University — SCALE Program (NASA Grant) | Nashville, TN
Space Applications Research Intern

December 2020 – May 2024

- Cut testing time **80%** by designing a microscope adapter and custom PCB with LabVIEW automation for radiation-effects research on silicon carbide semiconductors

PROJECTS

- **Amazon Robotics — Work-in-Progress Buffer (Senior Capstone, 2024)**: 4-conveyor work-in-progress buffer with computer vision to organize packages for a robotic sorting arm, projected to reduce manual exception handling by **75%**
- **Programmable Syringe Pump (2023)**: 3D CAD, custom electronics (Arduino, stepper drive, wiring), and firmware; flow from **1 μ L/min to 10 mL/min**

EDUCATION

Vanderbilt University | Nashville, TN
Bachelor of Engineering, Mechanical Engineering; Minor in Digital Fabrication

May 2024

SKILLS & CERTIFICATIONS

CAD & Analysis: SolidWorks (CSWA certified), Onshape, Fusion 360, Ansys, Altium Designer

Manufacturing: DFM for additive manufacturing, injection-molded part design (applying DfX feedback), rapid prototyping (3D printing, laser cutting, CNC), fixture design, first-article inspection

Electronics & Software: Custom PCB design (Altium), Arduino/ESP32 firmware, computer vision, MATLAB, Python, LabVIEW, workflow automation

Testing & Validation: Root cause analysis, supplier 8D review, acoustic/vibration/thermal testing, test fixture design

Languages: English (native), Spanish (intermediate)